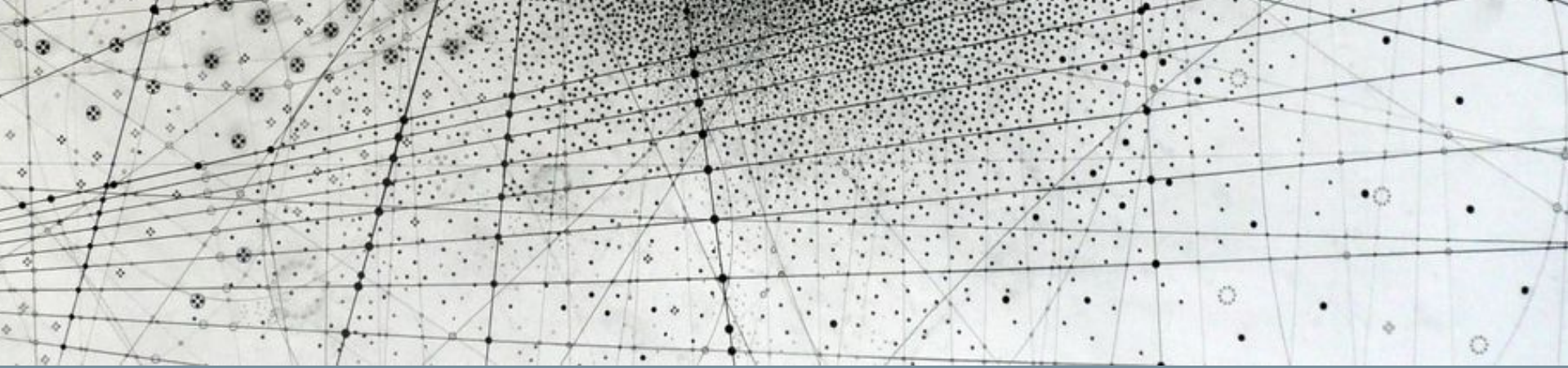




Research Methods in Machine Learning & Physics

Quick recap

Self-supervised learning (SSL)

Representation learning = learning representations of the data that make it easier to extract useful information when building classifiers or other predictors

Self-supervised learning (SSL) is a specific approach to representation learning in which you empower a model to define its own labels.

- It is the backbone of most large-scale foundation models today, e.g. LLMs and more
- This “pretext task” doesn’t matter on its own – instead, it serves to sculpt the embeddings
- Allows us to learn useful representations from unlabeled data

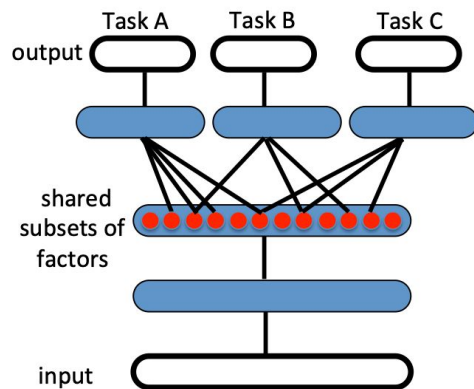


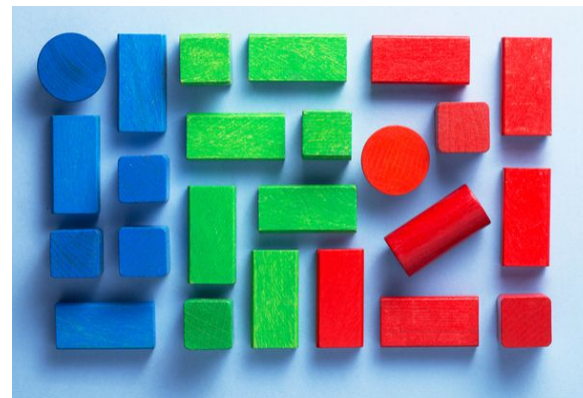
Fig. 1. Illustration of representation-learning discovering explanatory factors (middle hidden layer, in red), some explaining the input (semi-supervised setting), and some explaining target for each task. Because these subsets overlap, sharing of statistical strength helps generalization.

<https://arxiv.org/pdf/1206.5538>

Self-supervised learning (SSL)

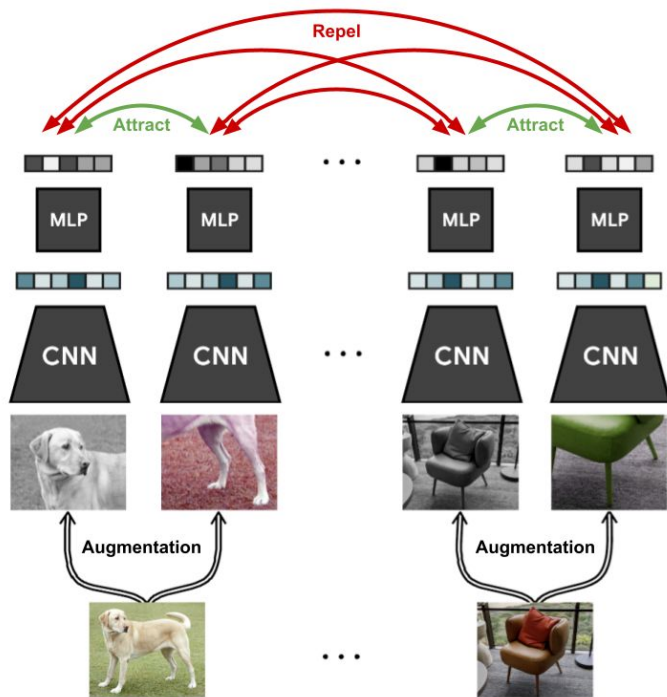
Types of SSL include **contrastive** & **non-contrastive** (e.g. generative).

Contrastive SSL means learning an embedding space that groups similar pairs of samples close to each other and pushes dissimilar ones far apart.



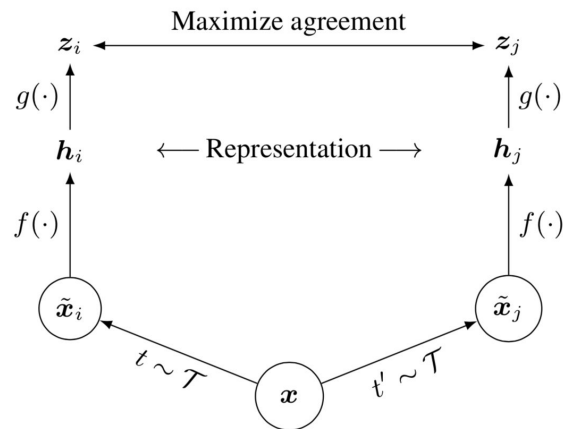
SimCLR

One canonical example of **contrastive SSL** is **SimCLR**, which uses a variant of the InfoNCE loss applied to different augmentations of the same input image.



A Simple Framework for Contrastive Learning of Visual Representations

Ting Chen¹ Simon Kornblith¹ Mohammad Norouzi¹ Geoffrey Hinton¹



Non-contrastive SSL, Part 1

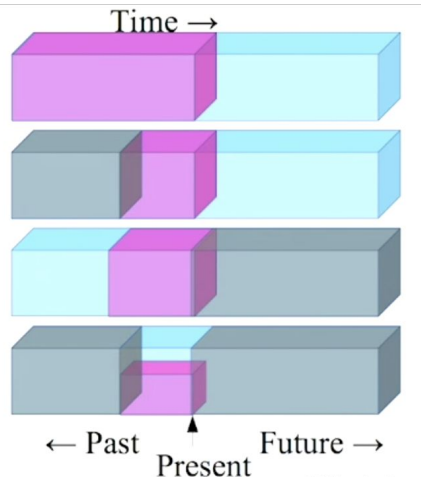
Non-contrastive SSL

What other choices do we have for pretext tasks, apart from augmenting & contrasting?

For instance, what if we don't know what kinds of augmentations are applicable or useful?

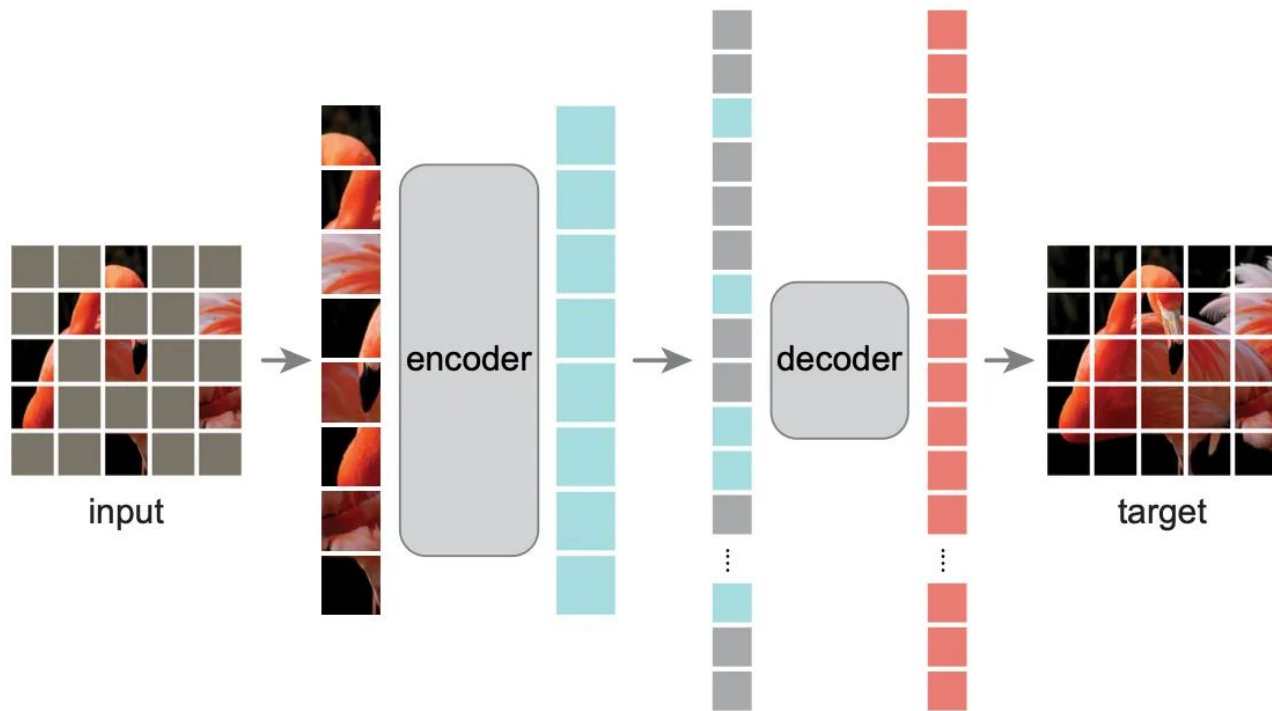
- What's an “augmentation” of a sentence?

- ▶ Predict any part of the input from any other part.
- ▶ Predict the **future** from the **past**.
- ▶ Predict the **future** from the **recent past**.
- ▶ Predict the **past** from the **present**.
- ▶ Predict the **top** from the **bottom**.
- ▶ Predict the occluded from the visible
- ▶ **Pretend there is a part of the input you don't know and predict that.**



Slide: LeCun

Generative SSL: Masked image modeling



Generative SSL: Masked image modeling

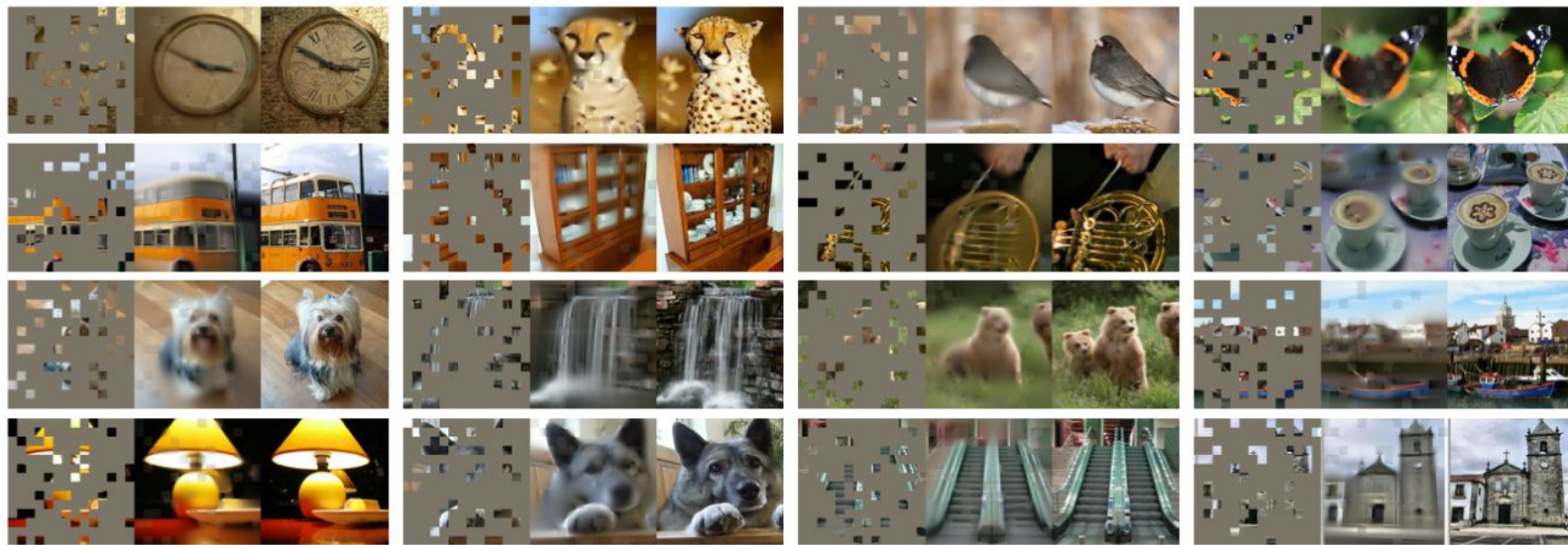
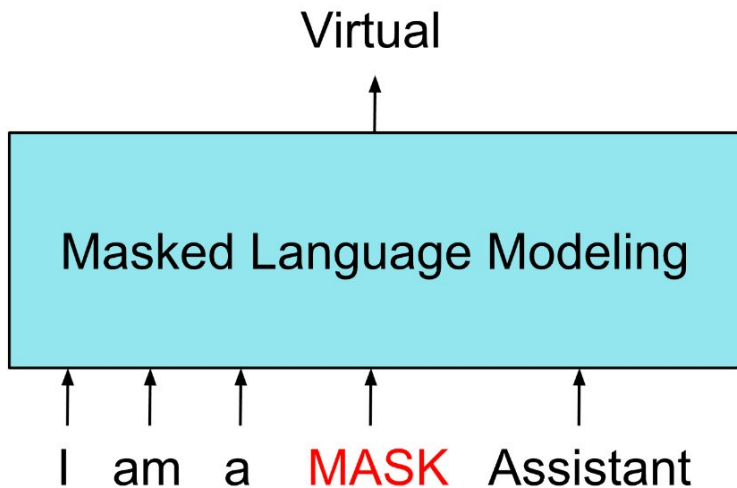


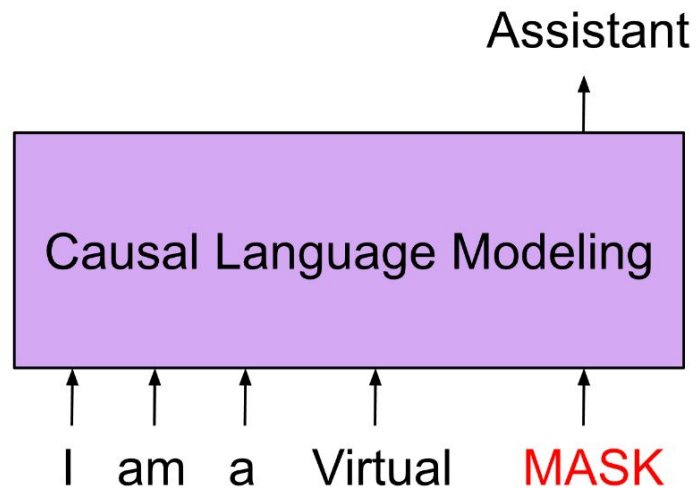
Figure 2. Example results on ImageNet *validation* images. For each triplet, we show the masked image (left), our MAE reconstruction[†] (middle), and the ground-truth (right). The masking ratio is 80%, leaving only 39 out of 196 patches. More examples are in the appendix.
[†]As no loss is computed on visible patches, the model output on visible patches is qualitatively worse. One can simply overlay the output with the visible patches to improve visual quality. We intentionally opt not to do this, so we can more comprehensively demonstrate the method's behavior.

Generative SSL: Masked language modeling



(a)

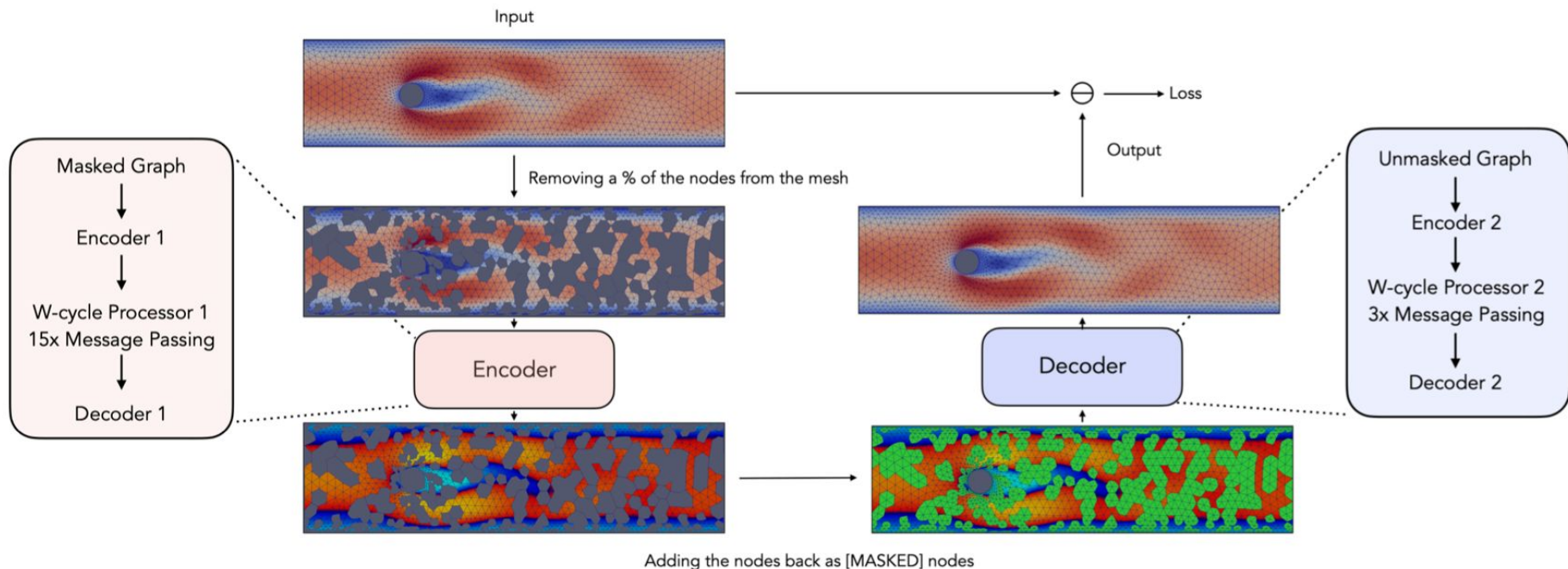
Not causal (e.g. BERT)



(b)

Causal/next-token (e.g. GPT)

Generative SSL: Masked modeling on physics data



Generative SSL: Masked modeling on physics data

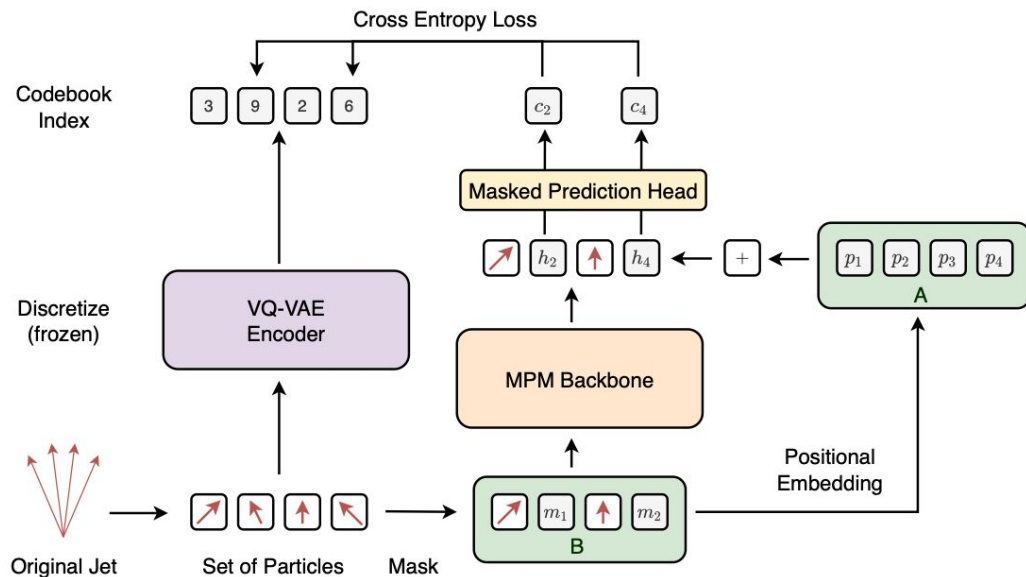


FIG. 1: The proposed model and training scheme for a FM for jets. A jet is represented as a set of particles, each a list of features, and some particles are replaced by a learnable vector and passed through a transformer encoder. Training aims to predict the discrete token identity, defined by the encoder of a pre-trained VQ-VAE, of the masked particles. The MPM Backbone in this work is a transformer encoder in this work. The positional embedding shown in box A is used only in the prediction head to preserve the permutation invariance in the backbone; it is required in the model to break the degeneracy from when $m_1 = m_2$.

In-class work

One person from each group will present their paper to the class:

1. An example of masked *language* modeling
2. An example of masked *image* modeling
3. An example of masked modeling on *physics data*

Add slides here: <https://tinyurl.com/phys805mm>